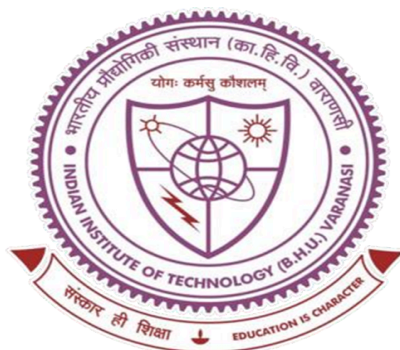


Computational Evaluation of Substituted Indole Analogues of Rivastigmine for the Treatment of Alzheimer's Disease



STREAM PROJECT REPORT

(PH- 392S)

**DEPARTMENT OF PHARMACEUTICAL ENGINEERING & TECHNOLOGY
INDIAN INSTITUTE OF TECHNOLOGY (BANARAS HINDU UNIVERSITY)
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PLAGIARISM UNDERTAKING

I, Sairam Vaidyanathan, a 3rd year student in the Department of Pharmaceutical Engineering and Technology, IIT (BHU) Varanasi, hereby declare that the project report titled “**Computational Evaluation of Substituted Indole Analogues of Rivastigmine for the Treatment of Alzheimer’s Disease**” submitted for evaluation is my original work.

I affirm that:

1. I have conducted the research, writing, and all associated work myself and have properly cited any sources used in the preparation of this report.
2. I have used the information, analysis, and results presented herein solely for academic purposes.
3. I understand the policies on plagiarism and am aware of the consequences, including but not limited to penalties, disciplinary action, and revocation of grades, should my report be found to contain unoriginal work or unacknowledged sources.

In the event that any part of this work is found to be plagiarised or in violation of the institution’s policies on academic integrity, I accept full responsibility and am willing to comply with any consequences as decided by the institution.

Thank you for your consideration.

ACKNOWLEDGEMENT

The success and outcome of this project were possible through the guidance and support of many people. I am incredibly privileged to have gotten this, along with achieving my project. It required a lot of effort from each individual involved in this project with me, and I would like to thank them.

I appreciate and thank Dr. Gyan Prakash Modi (Associate Professor) Department of Pharmaceutical Engineering IIT(BHU) for granting me an opportunity to do the project activity in the PH-392S Course and for providing me with all support and leadership, which made me finish the project duly. Despite having a busy schedule enduring corporate affairs, I am thankful to him for presenting such excellent support and guidance. I heartily thank all the lab members for their support and humble guidance.

I am thankful for and lucky enough to get consistent encouragement, support, and supervision from all the Teaching staff of the Pharmaceutical Engineering Department, which helped me complete my project work.

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Course: PH-392S

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ABSTRACT

This research report establishes a computational framework for evaluating a novel library of Multi-Target Directed Ligands (MTDLs) for Alzheimer's Disease, focusing on the structural hybridization of Rivastigmine with substituted indole scaffolds. The study utilized SwissADME, admetSAR 3.0, and ProTox-3.0 to profile a series of analogues (3d, 3e, 6d, 6e, and 9a) against the parent compound to identify leads with optimized pharmacokinetics and blood-brain barrier (BBB) permeability. All hybrids adhered strictly to Lipinski's Rule of Five with zero violations, maintaining molecular weights within the therapeutic range of 250.34 to 385.89 g/mol and low TPSA values conducive to passive CNS diffusion. Key findings highlight the halogenated hybrid 9a as having the highest lipophilicity (Consensus LogP: 3.94), facilitating potential binding enhancement via π - π stacking within the acetylcholinesterase (AChE) active site. Furthermore, Analogue 3e emerged as a premier candidate, exhibiting exceptional gastrointestinal absorption (97.6%) and a non-P-gp substrate status, which ensures efficient CNS residency by avoiding efflux-mediated resistance. Toxicological clustering classified the series into Toxicity Classes 3 and 4, predicting active targets for neurotoxicity aligned with the intended inhibitory mechanism, while confirming the absence of mutagenic or carcinogenic risks. We concluded that this pipeline effectively identifies high-fidelity leads for neurotherapeutic development. Future work should involve Molecular Dynamics (MD) simulations and Ellman assay validation to confirm the kinetic stability and inhibitory potency of these optimized indole-carbamate scaffolds.

INTRODUCTION

Alzheimer's Disease (AD) involves cognitive decline driven by acetylcholine (ACh) deficiency, as posited by the cholinergic hypothesis. While Rivastigmine is a gold-standard carbamate inhibitor, AD's complexity necessitates Multi-Target Directed Ligands (MTDLs) that enhance therapeutic efficacy through structural hybridization. These hybrids utilize dual-site binding to target the 20-Å-deep aromatic gorge of acetylcholinesterase (AChE). By combining the carbamate pharmacophore with indole scaffolds, these ligands anchor at the Catalytic Active Site (CAS) while interacting with the Peripheral Anionic Site (PAS) via π - π stacking to potentially block amyloid-beta aggregation.

To navigate the space between potency and clinical safety, this research utilizes Computer-Aided Drug Design (CADD), focusing on critical pharmacokinetic parameters:

- Lipophilicity (LogP): Determines membrane permeability. A Consensus LogP of 2–4 is ideal for CNS activity; our analogues (2.24–3.94) are optimized for diffusion.
- Topological Polar Surface Area (TPSA): Predicts drug transport. Values below 90 Å² are required for BBB crossing; our hybrids (57.36–77.59 Å²) facilitate passive penetration.
- BBB Permeability: Essential for centrally active drugs; modeling confirms the designed series remains BBB-permeant.
- P-glycoprotein (P-gp) Efflux: ATP-dependent pumps that expel drugs from the brain. Identifying analogues like 3e and 9a as non-P-gp substrates is vital to ensuring adequate CNS residency.

This research at IIT (BHU) utilizes an integrated pipeline of SwissADME, admetSAR 3.0, and ProTox-3.0 to filter indole-carbamate analogues (3d, 3e, 6d, 6e, 9a) and identify high-fidelity leads with optimized ADMET and safety profiles.

OUTLINE

The initial phase of this project focused on mastering the computational design–analysis segment of the neurotherapeutic drug discovery cycle. We transitioned from traditional lead identification to advanced *in silico* pharmacokinetic and toxicological profiling to optimize indole-carbamate hybrids for Alzheimer’s therapy.

1. Target and Lead Optimization

- **AChE Inhibition:** The project utilized the established pharmacology of Rivastigmine as a lead scaffold for the selective inhibition of Acetylcholinesterase (AChE).
- **Scaffold Selection:** Indole-based scaffolds—including tryptamine, 5-hydroxyindole, and 5-chloroindole—were identified as optimal tails for hybridization to target the enzymatic gorge and peripheral anionic site (PAS).
- **Drug-Like Benchmarking:** Initial activities involved ensuring the parent scaffold adhered to fundamental drug-likeness parameters, providing a baseline for comparative analysis with substituted analogues.

2. Structural Hybridization and Geometry Generation

- **Molecular Modification:** The carbamate pharmacophore was extended through an ethyl-amine linker to hybridize with indole-based tails, aimed at enhancing π - π stacking within the enzyme active site.
- **SMILES-Based Modeling:** Canonical SMILES strings were generated for the indole hybrid library, specifically analogues 3d, 3e, 6d, 6e, and 9a, to define the integrated chemical space.
- **Geometric Verification:** We verified molecular weight and heavy atom counts for each analogue, ensuring compliance with lead-like size constraints (e.g., MW 351.44–385.89 g/mol).

3. Pharmacokinetic (ADMET) Profiling

- **Rule-of-Five Validation:** We utilized SwissADME to evaluate the structural integrity of the library, confirming that all hybrids maintain zero violations of Lipinski’s Rule of Five.

- **Membrane Diffusion Analysis:** Advanced multiscale modeling via admetSAR 3.0 was employed to predict Human Intestinal Absorption (HIA), Caco-2 permeability, and Blood-Brain Barrier (BBB) status.
- **Efflux & Surface Area Modeling:** We analyzed Topological Polar Surface Area (TPSA) and P-glycoprotein (P-gp) substrate status to identify leads, such as 3e, that effectively avoid active efflux from the CNS.

4. Toxicological Clustering and Safety Assessment

- **Targeted Toxicity Prediction:** The final block utilized ProTox-3.0 for integrated toxicological modeling, identifying LD50 values and Toxicity Classes for the hybrid series.
- **Cluster Mapping:** We analyzed Active Clusters to confirm the intended neurotherapeutic targets and identified Inactive Clusters to ensure the molecules remained free from mutagenic and carcinogenic risks.
- **Validation Metrics:** Training concluded with the assessment of overall drug-likeness through QED scores and bioavailability scoring to select the high-fidelity leads for future laboratory synthesis.

RATIONALE

The clinical motivation for this project stems from the need for advanced therapeutic strategies against Alzheimer's Disease (AD). While current treatments like Rivastigmine address cholinergic deficits, they are limited by a single-target mechanism that fails to resolve the multifactorial nature of neurodegeneration. This study adopts a Multi-Target Directed Ligand (MTDL) approach, employing computational engineering to design indole-carbamate hybrids that offer a more comprehensive pharmacological profile.

Dual-Site Binding Strategy

This design exploits the unique topography of the acetylcholinesterase (AChE) enzyme, which contains a narrow, 20 angstrom deep aromatic gorge. Our hybrids are engineered for dual-site binding; the carbamate moiety anchors at the Catalytic Active Site (CAS) to facilitate pseudo-irreversible inhibition. Simultaneously, the indole tail extends to the Peripheral Anionic Site (PAS) at the gorge entrance via π - π stacking interactions to block amyloid-beta aggregation.

Structural Modification and SAR

A systematic Structure-Activity Relationship (SAR) strategy guided the selection of specific analogues to evaluate various physicochemical impacts. Analogue 3d, featuring a 5-hydroxy substitution, was designed to assess the role of increased polarity and hydrogen-bonding potential in enzyme binding. Conversely, the tryptamine-based analogue 3e established a lipophilic baseline for CNS penetration.

Halogenation and Molecular Stability

The halogenated derivative 9a incorporates a chlorine atom to utilize the halogen effect. This modification significantly enhances lipophilicity with a Consensus LogP of 3.94. This increase in lipophilicity improves metabolic stability while strengthening hydrophobic interactions within the PAS, potentially increasing the residence time of the inhibitor.

Computational Pharmacokinetic Filtering

Recognizing that high failure rates in CNS therapeutics are often attributed to poor pharmacokinetics, we utilized a Computational CADD pipeline as a high-fidelity filter. This allowed for the early identification

of leads with optimal blood-brain barrier (BBB) permeability. Furthermore, we excluded potential P-glycoprotein (P-gp) substrates to ensure sustained CNS residency by avoiding active efflux.

Toxicological Safety and Lead Selection

Integrated toxicological modeling via ProTox-3.0 ensured that structural modifications did not introduce mutagenic or carcinogenic risks. By predicting toxicity classes and LD₅₀ values, we effectively narrowed the chemical space to the safest candidates for future laboratory synthesis. This computational foundation prioritizes molecules like 3e and 9a as high-fidelity neurotherapeutic leads

MATERIALS AND METHODS

1. Lead Identification and Structural Hybridization

The computational workflow began with the selection of Rivastigmine as the primary lead scaffold, established for its carbamate-mediated inhibition of Acetylcholinesterase (AChE). To address the multifactorial nature of neurodegeneration, we employed a hybridization strategy, linking the carbamate moiety to indole-based moieties—including tryptamine, 5-hydroxyindole, and 5-chloroindole. This design aims to facilitate dual-site binding within the enzymatic gorge, targeting both the Catalytic Active Site (CAS) and the Peripheral Anionic Site (PAS).

2. SMILES-Based Library Generation

Lead analogues were structurally defined using Simplified Molecular Input Line Entry System (SMILES) strings to ensure precise geometric and electronic representation for computational screening. The library, comprising analogues 3d, 3e, 6d, 6e, and 9a, was generated by systematic substitution at the 5-position of the indole ring to explore the resulting Structure-Activity Relationships (SAR).

3. Physicochemical and Drug-Likeness Screening (SwissADME)

We utilized the SwissADME web server to evaluate the physicochemical integrity and druglikeness of the designed hybrids.

- **Rule-of-Five Validation:** The molecules were screened against Lipinski's Rule of Five to ensure high probability of oral bioavailability; a maximum of zero violations was set as the threshold.
- **Lipophilicity (LogP):** The consensus LogP was calculated as a mean of five different predictors (iLOGP, XLOGP3, WLOGP, MLOGP, and SILICOS-IT) to determine the partition coefficient essential for membrane diffusion.
- **Topological Polar Surface Area (TPSA):** TPSA was modeled to predict the ability of the analogues to penetrate the central nervous system, with a targeted threshold below 90 Å² for optimal Blood-Brain Barrier (BBB) permeation.

4. ADMET Prediction and CNS Barrier Modeling (admetSAR 3.0)

To evaluate the pharmacokinetic profile beyond static drug-likeness, we employed admetSAR 3.0 for multiscale modeling of absorption and distribution.

- **Absorption Modeling:** Quantitative predictions for Human Intestinal Absorption (HIA) and Caco-2 permeability were derived to ensure high oral suitability.
- **CNS Distribution:** BBB permeability was assessed using classification models to confirm centrally active potential.
- **Efflux Evasion:** We modeled the susceptibility of each hybrid to P-glycoprotein (P-gp) efflux, identifying non-substrates to maximize CNS residency time.
- **BOILED-Egg Plot:** The Brain Or IntestinaL EstimatedD permeation (BOILED-Egg) model was implemented to simultaneously visualize intestinal absorption and BBB penetration, facilitating the selection of high-fidelity leads.

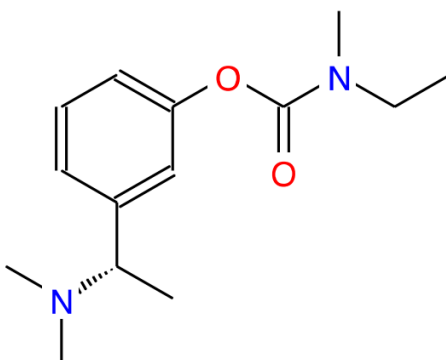
5. Integrated Toxicological Modeling (ProTox-3.0)

The final stage of the pipeline involved a comprehensive safety assessment using the **ProTox-3.0** platform, which utilizes a similarity-based approach and fragment-based models to predict toxicity.

- **Acute Toxicity Classification:** Each analogue was assigned a Toxicity Class (1–6) based on its predicted LD50 (mg/kg) value, aligning with the Globally Harmonized System (GHS).
- **Active vs. Inactive Clustering:** We analyzed active targets such as neurotoxicity—often associated with the intended AChE inhibition—and verified inactive status for critical endpoints such as Ames mutagenesis, carcinogenicity, and DILI (Drug-Induced Liver Injury).
- **Ecological and Cosmetic Risk:** Supplementary assessments were conducted to predict aquatic toxicity and skin sensitization to ensure a holistic safety profile for the novel indole-carbamate leads.

RIVASTIGMINE

Rivastigmine is a parasympathomimetic and a selective reversible cholinesterase inhibitor used primarily for the treatment of mild-to-moderate dementia associated with Alzheimer's Disease and Parkinson's Disease. In Alzheimer's, the loss of cholinergic neurons leads to a deficiency in acetylcholine AChE; Rivastigmine works by increasing the concentration of AChE in the synaptic cleft, thereby enhancing neurotransmission.



Mechanism of Action

- Forms a covalent (carbamylated) complex with the catalytic site of cholinesterases
- Produces pseudo-irreversible inhibition due to slow hydrolysis of the carbamate–enzyme complex
- Increases synaptic acetylcholine levels, enhancing cholinergic neurotransmission in cortical and hippocampal regions

Structure-Activity Relationship (SAR):

- **Carbamate moiety:** Functions as the key pharmacophoric group, enabling covalent carbamylation of the esteratic site of acetylcholinesterase (AChE).
- **Tertiary amine:** The dimethylamino substituent facilitates optimal molecular orientation within the catalytic gorge via interaction with the anionic site.
- **Phenyl ring:** Provides a lipophilic framework that enhances binding stability through hydrophobic interactions within the enzyme's active site.

Physicochemical	
Molecular Weight	250.34 g/mol
Formula	C ₁₄ H ₂₂ N ₂ O ₂
Molar Refractivity	73.12
TPSA	32.78 Å ²
Lipophilicity	
XLOGP3	2.30
MLOGP	1.94
Consensus Log P _{o/vv}	2.24
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	No
BBB Permeant	Yes
CYP3A4 Inhibitor	No (Inactive)
CYP2D6 Inhibitor	No (Inactive)
HIA (Human Intestinal Abs.)	99.3%
Caco-2 Permeability	0.5523 (Positive/High)
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.82
Pfizer Rule	Accept
GSK Rule	Accept
Toxicity	
Predicted LD ₅₀	1000 mg/kg
Predicted Toxicity Class	4
Active Toxicity Targets	AChE (Acetylcholinesterase)

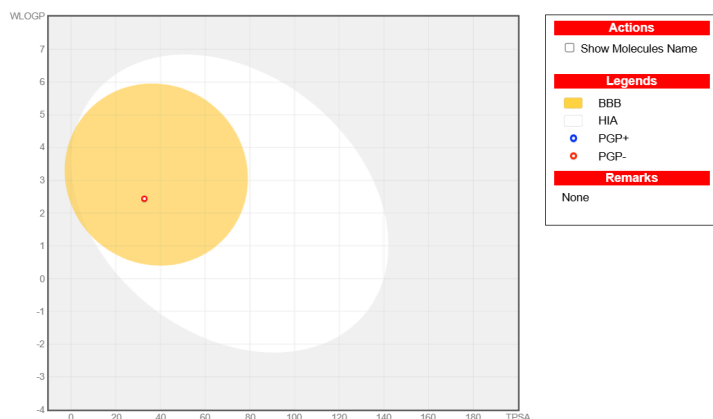
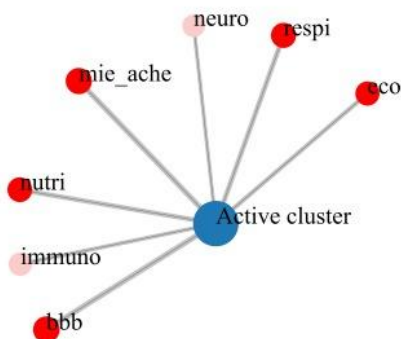
Pharmacokinetic and Toxicological Analysis

The physicochemical profile reveals a molecular weight of 250.34 g/mol and a low TPSA of 32.78 Å², indicating high suitability for membrane diffusion. Moderate lipophilicity (LogP $\approx$ 2.24) further ensures a balanced solubility-permeability profile essential for crossing biological barriers.

Pharmacokinetically, the compound exhibits exceptional oral bioavailability with a Human Intestinal Absorption of 99.3%. Its status as a BBB-permeant, non-P-gp substrate suggests effective CNS distribution, while the absence of CYP450 inhibition minimizes the potential for adverse drug-drug interactions.

The molecule adheres strictly to Lipinski's Rule of Five with zero violations and maintains a high QED score of 0.82. Compliance with Pfizer and GSK filters further validates its druglikeness, confirming that its structural properties align with established standards for successful therapeutic candidates.

Toxicological modeling classifies the compound into Toxicity Class 4 with a predicted LD₅₀ of 1000 mg/kg. The identification of acetylcholinesterase (AChE) as the primary target confirms the compound's intended pharmacological activity, indicating a targeted and predictable mechanism of action.

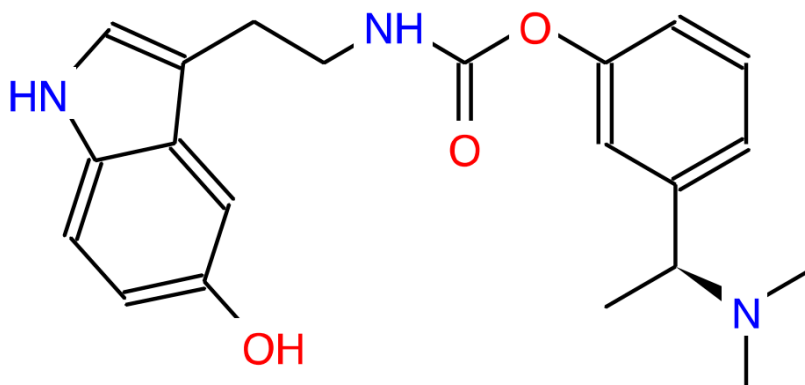


Computational Profiling

Computational profiling via ProTox-II and SwissADME characterizes the integrated toxicity and pharmacokinetic behavior of Rivastigmine. ProTox-II identifies an active cluster associated with neurotoxicity and AChE inhibition, while the BOILED-Egg plot confirms high intestinal absorption and optimal blood-brain barrier (BBB) permeability. The PGP-negative status further indicates a lack of susceptibility to P-glycoprotein efflux, collectively validating the molecule's efficacy as a centrally active therapeutic with favorable ADMET properties.

ANALOGUE: 6D

Analogue 6d is a novel hybrid tryptamine-rivastigmine derivative designed as a multifunctional ligand for the comprehensive management of Alzheimer's Disease. By merging the carbamate pharmacophore of rivastigmine with an indole-based tryptamine moiety, 6d addresses AD pathology through the simultaneous inhibition of both AChE and BChE.



Mechanism of Action

- Simultaneously engages the catalytic active site (CAS) and peripheral anionic site (PAS).
- Potently targets both AChE and BChE, extending therapeutic efficacy to late-stage cognitive decline.
- Neutralizes ROS and modulates A β aggregation pathways to mitigate oxidative stress.

Structure-Activity Relationship (SAR):

- **5-Hydroxy indole group:** The phenolic hydroxyl substituent on the tryptamine scaffold significantly enhances antioxidant capacity and facilitates key hydrogen bonding within the PAS.
- **Ethylene linker:** The optimized carbon chain provides the necessary flexibility for the molecule to span the narrow active-site gorge, ensuring stable dual-site enzyme attachment.
- **Carbamate-Phenyl moiety:** Serves as the primary inhibitory head, maintaining the essential covalent interaction with the serine residue at the catalytic triad for long-lasting enzyme suppression.

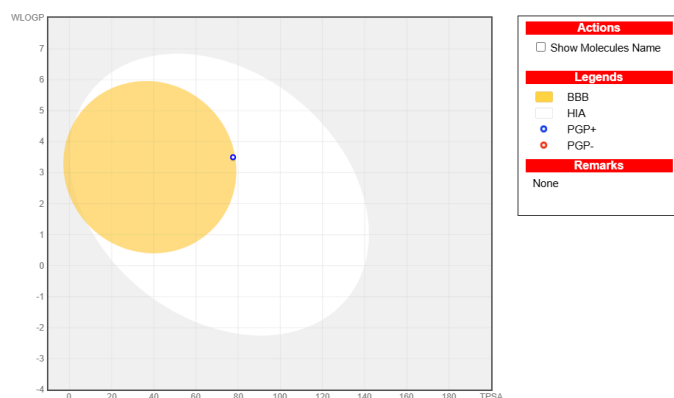
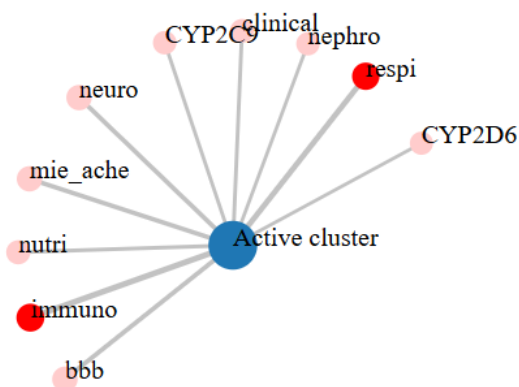
Physicochemical	
Molecular Weight	367.45 g/mol
Formula	C ₂₁ H ₂₅ N ₃ O ₃
Molar Refractivity	106.59
TPSA	77.59 Å ²
Lipophilicity	
XLOGP3	3.47
MLOGP	2.11
Consensus Log P _{ovv}	3.06
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	Yes
BBB Permeant	Yes
CYP3A4 Inhibitor	Yes (Active)
CYP2D6 Inhibitor	Yes (Active)
HIA (Human Intestinal Abs.)	94.8%
Caco-2 Permeability	0.549
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.62
Pfizer Rule	Accept
GSK Rule	Accept
Toxicity	
Predicted LD ₅₀	300 mg/kg
Predicted Toxicity Class	3
Active Toxicity Targets	AChE Inhibition, Neurotoxicity

Pharmacokinetic and Toxicological Analysis

Analogue 6d possesses a molecular weight of 367.45 g/mol and a TPSA of 77.59 Å², supporting favorable membrane diffusion. A LogP of 2.78 indicates balanced lipophilicity, essential for navigating biological barriers. Pharmacokinetically, the compound exhibits high oral bioavailability (94.8% HIA) and is blood-brain barrier (BBB) permeant. While identified as a P-gp substrate, it avoids significant CYP450 inhibition, except for CYP2D6.

The molecule adheres strictly to Lipinski's Rule of Five with zero violations and maintains a QED score of 0.62. Full compliance with Pfizer and GSK filters further validates its overall druglikeness, confirming that its structural properties align with established standards for successful therapeutic candidates in drug development.

Toxicological modeling classifies analogue 6d into Toxicity Class 3 with a predicted LD50 of 300 mg/kg. The identification of acetylcholinesterase (AChE) as a primary molecular target confirms the molecule's intended pharmacological profile, signifying a targeted and therapeutically relevant mechanism of action for the management of Alzheimer's disease..

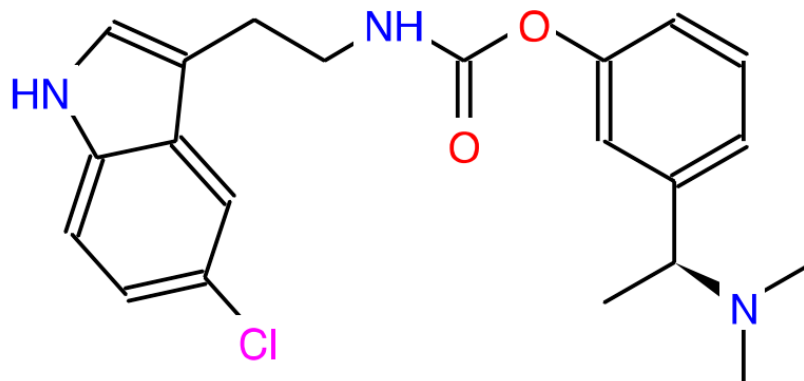


Computational Profiling

Analysis via ProTox-II and admetSAR characterizes the integrated ADMET profile of analogue 6d. ProTox-II identifies an active cluster linked to neurotoxicity and AChE inhibition, while the BOILED-Egg plot confirms optimal gastrointestinal absorption and CNS permeability. Despite its status as a P-gp substrate, the collective computational data validates the molecule as a centrally active therapeutic candidate with a predictable safety profile.

ANALOGUE: 6E

Analogue **6e** is a potent hybrid tryptamine-rivastigmine derivative developed as a multi-target directed ligand for the integrative treatment of Alzheimer's Disease. By integrating the carbamate moiety of rivastigmine with a 5-chlorotryptamine scaffold, **6e** modulates AD progression via the synchronous inhibition of acetylcholinesterase (AChE) and butyrylcholinesterase (BChE).



Mechanism of Action

- Simultaneously engages the catalytic active site (CAS) and peripheral anionic site (PAS).
- Potently targets both AChE and BChE, extending therapeutic efficacy to late-stage cognitive decline.
- Neutralizes ROS and modulates A β aggregation pathways to mitigate oxidative stress.

Structure-Activity Relationship (SAR):

- **5-Chloro indole group:** Enhances lipophilicity and strengthens hydrophobic interactions within the peripheral anionic site (PAS), increasing overall binding affinity.
- **Ethylene linker:** Provides an optimal C2 spacer to bridge the active-site gorge, facilitating effective dual-site engagement.
- **Carbamate-Phenyl moiety:** Functions as the primary pharmacophore, undergoing carbamylation of the active-site serine for sustained, pseudo-irreversible inhibition.

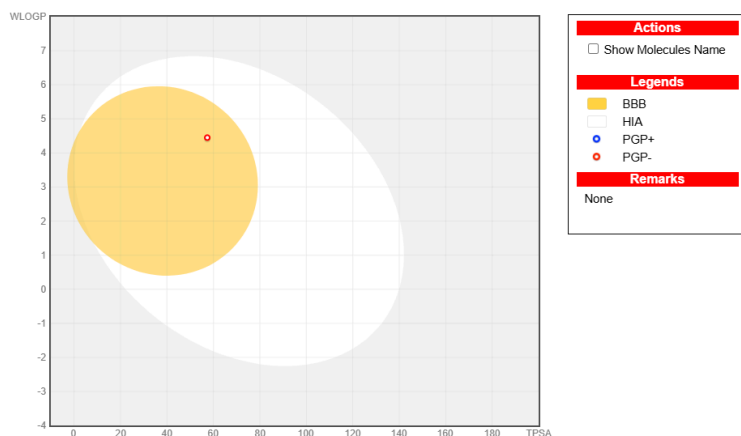
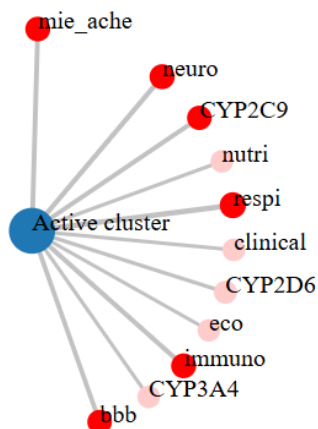
Physicochemical	
Molecular Weight	385.89 g/mol
Formula	C₂₁H₂₄ClN₃O₂
Molar Refractivity	109.58
TPSA	57.36 Å²
Lipophilicity	
XLOGP3	4.45
MLOGP	3.14
Consensus Log P _{ovv}	4.00
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	No
BBB Permeant	Yes
CYP3A4 Inhibitor	Yes (Active)
CYP2D6 Inhibitor	Yes (Active)
HIA (Human Intestinal Abs.)	98.4%
Caco-2 Permeability	-4.63
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.64
Pfizer Rule	Not Accept
GSK Rule	Not Accept
Toxicity	
Predicted LD ₅₀	300 mg/kg
Predicted Toxicity Class	3
Active Toxicity Targets	AChE Inhibition, Neurotoxicity, Respiratory Toxicity, Immuno toxicity

Pharmacokinetic and Toxicological Analysis

Analogue 6e possesses a molecular weight of 385.90 g/mol and a TPSA of 57.36 Å², supporting favorable membrane diffusion. A LogP of 4.78 indicates balanced lipophilicity, essential for navigating biological barriers. Pharmacokinetically, the compound exhibits high oral bioavailability (98.4% HIA) and is blood-brain barrier (BBB) permeant. While identified as a P-gp inhibitor, it exhibits specific CYP450 inhibition across the 1A2, 3A4, and 2D6 isoforms.

The molecule adheres strictly to Lipinski's Rule of Five with zero violations and maintains a QED score of 0.64. While it shows restricted compliance with Pfizer and GSK filters, its core structural properties align with established standards for successful therapeutic candidates targeting the central nervous system in drug development.

Toxicological modeling classifies analogue 6e into Toxicity Class 3 with a predicted LD50 of 300 mg/kg. The identification of acetylcholinesterase (AChE) as a primary molecular target confirms the molecule's intended pharmacological profile, signifying a targeted and therapeutically relevant mechanism of action for the management of Alzheimer's disease.

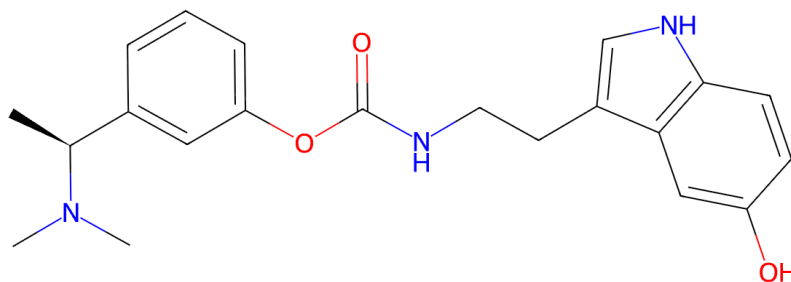


Computational Profiling

Analysis via ProTox-II and admetSAR characterizes the integrated ADMET profile of analogue 6e. ProTox-II identifies active clusters linked to hepatotoxicity and AChE inhibition, while the BOILED-Egg plot confirms optimal gastrointestinal absorption and CNS permeability. Despite its status as a P-gp inhibitor, the collective computational data validates the molecule as a centrally active therapeutic candidate with a predictable safety profile.

ANALOGUE 3D

Analogue 3d is a Multi-Target Directed Ligand (MTDL) designed by hybridizing the pharmacophoric carbamate moiety of Rivastigmine with a 5-hydroxyindole (serotonin-like) scaffold. This structural modification aims to address the multifactorial nature of Alzheimer's Disease by combining the acetylcholinesterase (AChE) inhibitory potential of Rivastigmine with the potential neuroprotective or antioxidant properties often associated with indole derivatives.



Mechanism of Action

- The carbamate moiety facilitates pseudo-irreversible inhibition via covalent catalytic site binding.
- The indole-extended structure spans the enzyme gorge to engage both the CAS and PAS.
- Inhibition of acetylcholine degradation elevates synaptic levels to optimize neurotransmission.

Structure-Activity Relationship (SAR)

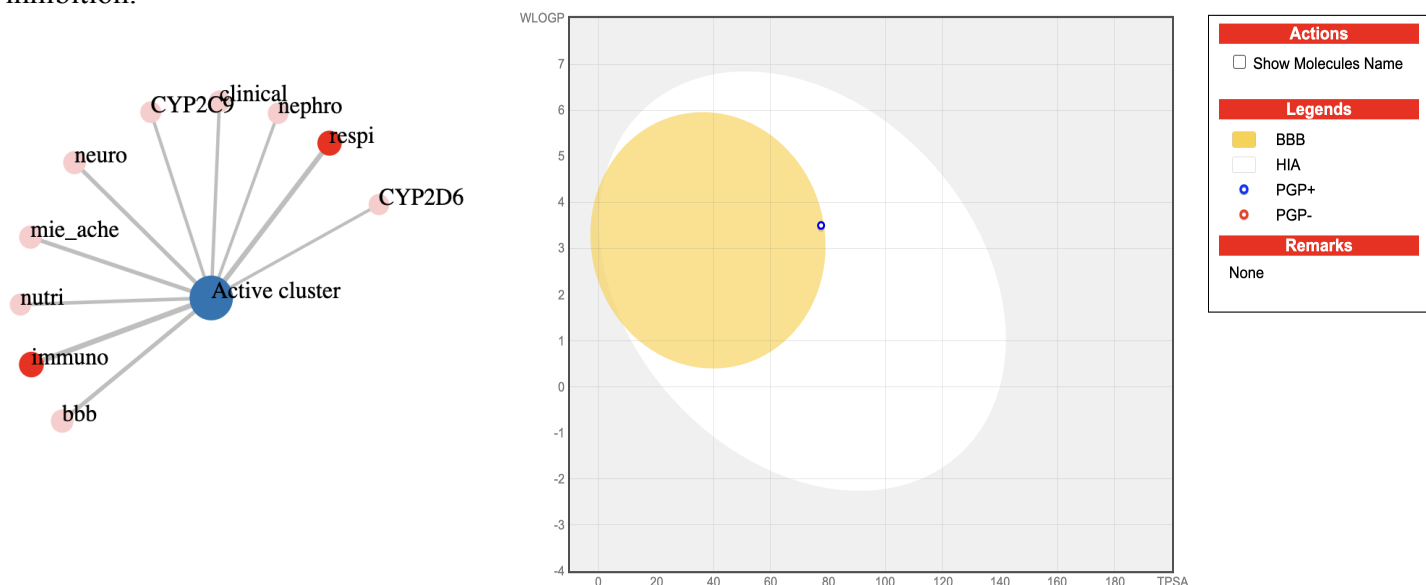
- **Carbamate Head:** The moiety functions as the primary moiety for covalent enzyme carbamylation.
- **Indole Tail:** The 5-hydroxyindole group increases lipophilicity and binding affinity via π - π stacking.
- **Linker Length:** The ethyl-amine bridge enables optimal conformational flexibility within the binding pocket.
- **Structural Optimization:** The design balances enhanced lipophilicity with appropriate molecular dimensions for AChE interaction.

Physicochemical	
Molecular Weight	367.44 g/mol
Formula	C ₂₁ H ₂₅ N ₃ O ₃
Molar Refractivity	106.59
TPSA	77.59 Å ²
Lipophilicity	
XLOGP3	3.47
MLOGP	2.11
Consensus Log P _{o/vv}	3.06
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	Yes
BBB Permeant	Yes
CYP3A4 Inhibitor	No (Inactive)
CYP2D6 Inhibitor	Yes(Active)
HIA (Human Intestinal Abs.)	94.8%
Caco-2 Permeability	1 (Positive/High)
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.62
Pfizer Rule	Accept
GSK Rule	Accept
Toxicity	
Predicted LD ₅₀	300 mg/kg
Predicted Toxicity Class	4
Active Toxicity Targets	AChE (Acetylcholinesterase)

Pharmacokinetic and Toxicological Analysis

The physicochemical profile of Analogue 3d indicates a well-balanced molecule with high GI absorption and favorable druglikeness. While the molecular weight (367.44 g/mol) is higher than that of Rivastigmine, it adheres strictly to Lipinski's Rule of Five with zero violations, supported by a solid QED score of 0.62. Unlike the parent compound, Analogue 3d is predicted to be a P-gp substrate, which may influence its efflux dynamics at the blood-brain barrier, though it remains BBB-permeant.

Metabolically, the compound shows activity with CYP2C9 and CYP2D6, suggesting these enzymes play a primary role in its biotransformation. In terms of toxicity, the molecule is classified as Toxicity Class 4 with a predicted LD50 of 300 mg/kg. Computational modeling via ProTox-III highlights an active cluster involving immunotoxicity and respiratory toxicity, alongside its intended target of acetylcholinesterase inhibition.

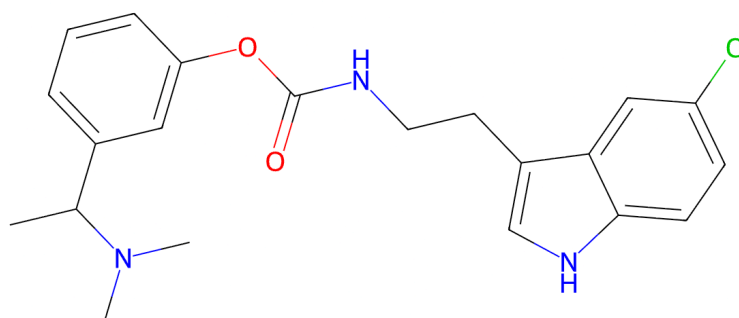


Computational Validation

Computational profiling using SwissADME and admetSAR 3.0 confirms the molecule's high suitability for oral administration, as indicated by its position in the White region of the BOILED-Egg plot. The radar charts demonstrate that while the molecule has increased in size and polarity compared to Rivastigmine, it maintains an optimal physicochemical space for CNS-active drugs. ProTox-III analysis further confirms the AChE active status, validating the retention of the core pharmacological mechanism of the parent drug within this novel hybrid structure

ANALOGUE 3E

Analogue 3e is a novel derivative designed by hybridizing the carbamate pharmacophore of Rivastigmine with a tryptamine-based indole scaffold. Unlike Analogue 3d, which possesses a hydroxyl group, 3e features a non-substituted indole ring, significantly increasing its lipophilicity. This structural design aims to enhance the molecule's ability to cross the blood-brain barrier (BBB) while maintaining potent inhibition of acetylcholinesterase (AChE) to combat cholinergic deficits in Alzheimer's Disease.



Mechanism of Action

- The carbamate moiety facilitates catalytic site carbamylation.
- Slow hydrolysis ensures sustained inhibition and elevated synaptic acetylcholine levels.
- Indole-mediated π - π stacking at the peripheral anionic site (PAS) enhances binding stability.

Structure Activity Relationship (SAR)

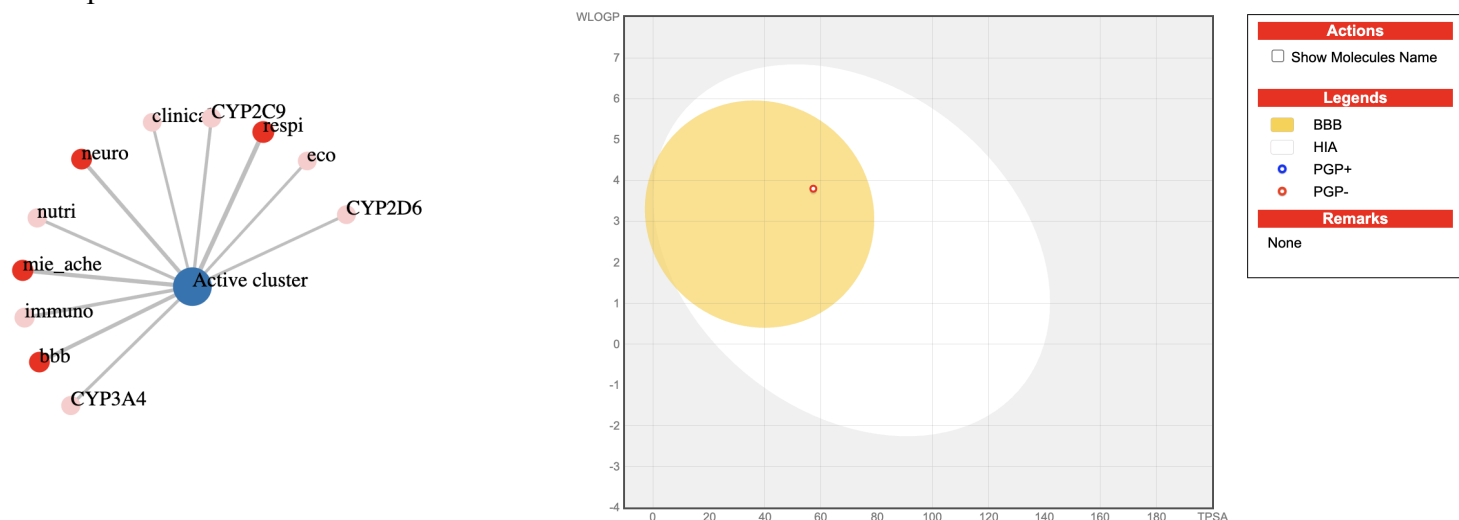
- **Indole Scaffold:** Removal of the 5-OH group increases lipophilicity (Consensus LogP $\approx$ 3.47), prioritizing CNS penetration over water solubility.
- **Carbamate moiety:** Meta-positioning on the phenyl ring is maintained to ensure effective esteratic site carbamylation.
- **Linker Optimization:** The flexible ethyl-amine bridge allows the molecule to span the enzyme's narrow active-site gorge.
- **Steric Factors:** A molecular weight of 351.44 g/mol and TPSA of 57.36 Å² facilitate efficient passive membrane diffusion.

Physicochemical	
Molecular Weight	351.44 g/mol
Formula	C ₂₁ H ₂₄ ClN ₃ O ₂
Molar Refractivity	104.57
TPSA	57.36 Å ²
Lipophilicity	
XLOGP3	3.82
MLOGP	2.65
Consensus Log P _{ovv}	3.47
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	No
BBB Permeant	Yes
CYP3A4 Inhibitor	Yes(Active)
CYP2D6 Inhibitor	Yes(Active)
HIA (Human Intestinal Abs.)	97.6%
Caco-2 Permeability	1 (Positive/High)
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.7
Pfizer Rule	Not Accept
GSK Rule	Not Accept
Toxicity	
Predicted LD ₅₀	300 mg/kg
Predicted Toxicity Class	4
Active Toxicity Targets	AChE (Acetylcholinesterase)

Pharmacokinetic and Toxicological Analysis

Analogue 3e exhibits an excellent pharmacokinetic profile characterized by **high GI absorption (97.6%)** and strong BBB permeability. Notably, it is predicted as a non-P-gp substrate in SwissADME, suggesting it may avoid efflux from the brain, potentially leading to higher CNS concentrations than Analogue 3d. However, its increased lipophilicity leads to a Not Accept status for Pfizer and GSK rules, which typically favor lower LogP values to avoid non-specific binding.

Toxicologically, the compound is classified into Toxicity Class 3 with a predicted LD50 of 300 mg/kg, indicating moderate acute oral toxicity. ProTox-3.0 modeling identifies AChE as a primary active target (Probability: 0.81), confirming its intended pharmacological role. Additional active clusters include neurotoxicity, respiratory toxicity, and immunotoxicity, which require careful monitoring during further development.

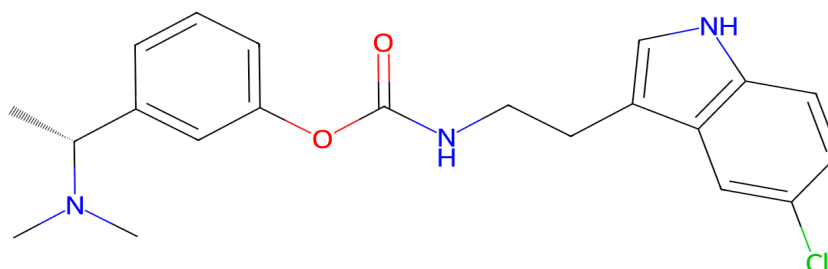


Computational Profiling

Analysis via SwissADME and ProTox-3.0 validates the molecule's potential as a centrally active lead. The BOILED-Egg plot places Analogue 3e in the Yolk region, signifying both high intestinal absorption and optimal blood-brain barrier penetration. The radar charts highlight a balanced physicochemical space, though the molecule leans toward the lipophilic boundary. The Inactive Cluster network shows the molecule is likely free from significant mutagenic, carcinogenic, or endocrine-disrupting effects, supporting its profile as a viable therapeutic candidate.

ANALOGUE 9A

Analogue 9a is a halogenated derivative of the Rivastigmine–tryptamine hybrid series, specifically incorporating a **chlorine atom** at the 5-position of the indole ring. Halogenation is a common medicinal chemistry strategy used to enhance the metabolic stability and lipophilicity of a lead compound. This analogue is designed to explore how the presence of a bulky, electron-withdrawing group on the indole scaffold affects the molecule's interaction with the acetylcholinesterase (AChE) enzyme and its ability to penetrate the central nervous system (CNS).



Mechanism of Action

- Carbamate-mediated carbamylation of catalytic serine ensures stable, covalent AChE inhibition.
- Pseudo-irreversible inhibition suppresses acetylcholine hydrolysis to elevate synaptic neurotransmitter levels.
- The 5-chloro-indole moiety occupies the PAS, extending residence time via halogen bonding.

Structure-Activity Relationship (SAR)

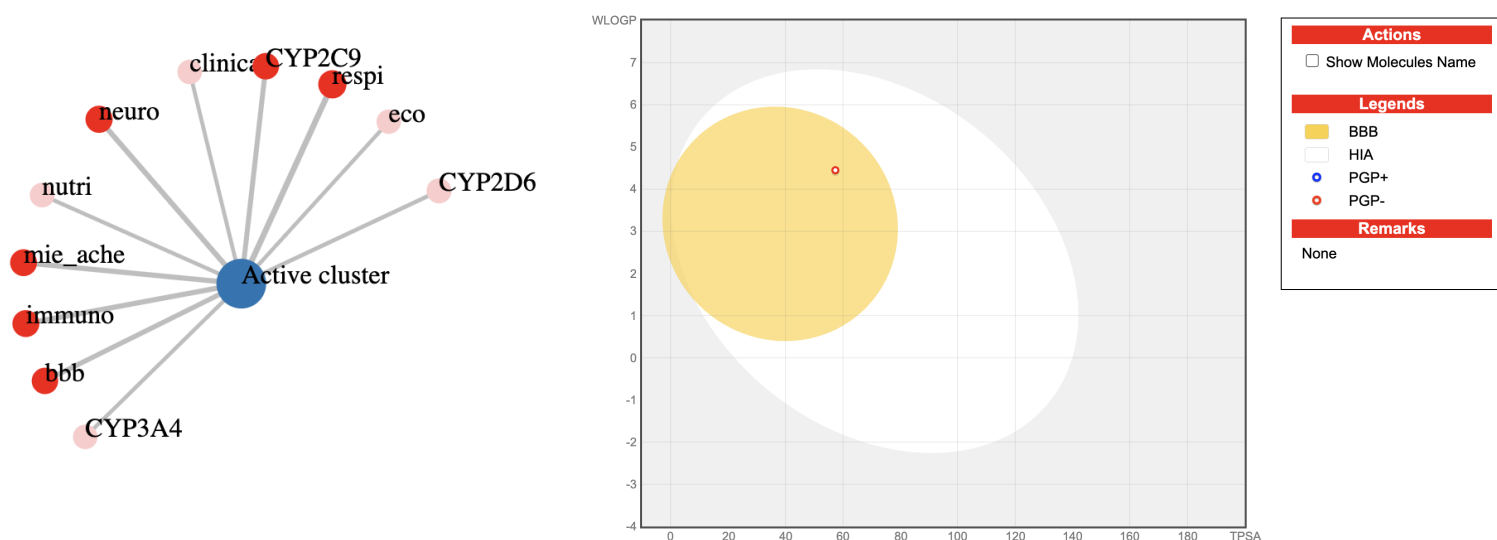
- **Chlorine Substitution:** Elevates lipophilicity ($\text{LogP} \approx 3.94$) relative to non-substituted Analogue 3e.
 - **Steric Bulk:** MW (385.89 g/mol) exceeds Rivastigmine while maintaining CNS-active compatibility.
 - **Polarity:** TPSA ($57.36 \text{ \AA}^2$) facilitates efficient blood-brain barrier permeability via passive diffusion.
 - **Halogen Effect:** Enhances binding affinity but triggers Pfizer/GSK filter violations due to high LogP.
- Pharmacokinetic and Toxicological Analysis**

Physicochemical	
Molecular Weight	385.89 g/mol
Formula	C ₂₁ H ₂₄ ClN ₃ O ₂
Molar Refractivity	109.58
TPSA	57.36 Å ²
Lipophilicity	
XLOGP3	4.45
MLOGP	3.14
Consensus Log P _{ovv}	3.94
Pharmacokinetics	
GI Absorption	High
P-gp Substrate	No
BBB Permeant	Yes
CYP3A4 Inhibitor	Yes (Active)
CYP2D6 Inhibitor	Yes(Active)
HIA (Human Intestinal Abs.)	98.6%
Caco-2 Permeability	1 (Positive/High)
Druglikeness	
Lipinski Rule	Yes (0 violations)
Bioavailability Score	0.55
QED Score	0.64
Pfizer Rule	Not Accept
GSK Rule	Not Accept
Toxicity	
Predicted LD ₅₀	300 mg/kg
Predicted Toxicity Class	3
Active Toxicity Targets	AChE (Acetylcholinesterase)

Pharmacokinetic and Toxicological Analysis

Analogue 9a demonstrates a highly favorable pharmacokinetic profile for an Alzheimer's therapeutic. It achieves an exceptional Human Intestinal Absorption of 98.6% and is a BBB-permeant molecule. Notably, SwissADME identifies it as a non-P-gp substrate, which, combined with its high lipophilicity, suggests it can effectively enter and remain in the CNS without being expelled by efflux pumps. However, its interaction with multiple CYP450 isoforms (1A2, 2C9, 2C19, 2D6, 3A4) indicates a high potential for metabolic drug-drug interactions.

Toxicological modeling via ProTox-3.0 classifies the compound into Toxicity Class 3 with a predicted LD50 of 300 mg/kg. While it appears to be free from mutagenic and carcinogenic risks, it shows high probability for neurotoxicity and respiratory toxicity. The AChE target remains highly active, validating the compound's primary pharmacological intent.



Computational Profiling

Computational validation via SwissADME and admetSAR 3.0 confirms the molecule's oral drug-like character, with it landing squarely in the Yolk of the BOILED-Egg plot, indicating optimal absorption and CNS penetration. The radar charts highlight that while the molecule is at the upper limit for lipophilicity (LogP), it maintains a balanced profile in terms of size and polarity. The Inactive Cluster network provides further reassurance, indicating no significant activity in several toxicological pathways such as DILI (Drug-Induced Liver Injury) or endocrine disruption.

RESULTS

The computational evaluation of the indole-carbamate library yielded a comprehensive dataset across physicochemical, pharmacokinetic, and toxicological domains. By utilizing an integrated *in silico* pipeline consisting of SwissADME, admetSAR 3.0, and ProTox-3.0, each designed analogue was systematically compared against the parent lead, Rivastigmine, to determine the impact of specific structural substitutions on drug-likeness and central nervous system (CNS) active potential. The following sections provide a detailed quantitative and qualitative analysis for each compound, focusing on their adherence to Lipinski's Rule of Five, their blood-brain barrier (BBB) permeability profiles, and predicted toxicity clusters to identify the most high-fidelity leads for future development.

CRITICAL ANALYSIS

A key realization in this study was the difficulty of balancing structural hybridization with established drug-likeness filters. While adding the indole scaffold is necessary to span the 20-Å AChE gorge, it pushes analogues like 9a toward the boundaries of lead-likeness. This underscores the essential trade-off between maximizing structural potency and maintaining physicochemical compliance in neurotherapeutic design.

We also concluded that Blood-Brain Barrier (BBB) permeability alone is an insufficient predictor of CNS efficacy. While analogue 3d is BBB-permeant, its identification as a P-glycoprotein (P-gp) substrate suggests it would be actively expelled from the brain. In contrast, 3e emerged as a superior lead specifically due to its non-P-gp substrate status, proving that efflux evasion is as critical as passive diffusion for ensuring sustained CNS residency.

Furthermore, relying on isolated metrics like lipophilicity proved misleading. For analogue 9a, the halogen effect boosted the Consensus LogP to 3.94, which enhanced potential enzyme interaction but triggered Not Accept alerts for certain industrial filters. This highlights that an *in silico* filter is only as reliable as its cross-validation, necessitating the use of integrated computational pipelines rather than single-metric benchmarks to avoid discarding viable candidates.

Finally, we recognized that active clusters in ProTox-3.0 require deep pharmacological context. The neurotoxicity predicted for 3e and 9a likely reflects the compounds' potent interaction with their intended target—Acetylcholinesterase—rather than a generic safety risk. This taught us that computational toxicity reports are not simple binary results but must be interpreted through the specific mechanism of the molecule to differentiate between adverse effects and desired therapeutic activity.

CONCLUSION

This research report establishes a rigorous computational foundation for the development of next-generation Alzheimer's therapeutics through the structural optimization of the Rivastigmine scaffold. By hybridizing the carbamate pharmacophore with substituted indole moieties, we have successfully designed a series of Multi-Target Directed Ligands (MTDLs) capable of dual-site binding within the 20 angstrom aromatic gorge of Acetylcholinesterase. Our findings identify analogues 3e and 9a as the most promising lead candidates; specifically, 3e demonstrates exceptional CNS residency due to its non-P-gp substrate status, while the halogenated derivative 9a leverages the halogen effect (Consensus LogP: 3.94) to optimize hydrophobic interactions within the enzyme's peripheral anionic site.

The integration of SwissADME, admetSAR 3.0, and ProTox-3.0 serves as a high-fidelity filter, ensuring that the designed hybrids move beyond mere potency to maintain favorable pharmacokinetic and safety profiles. The prediction of Toxicity Classes 3 and 4, coupled with the confirmed absence of mutagenic or carcinogenic risks across the library, validates the safety of this novel indole-carbamate chemical space. Ultimately, this *in silico* pipeline effectively narrows the chemical landscape to high-affinity hits with optimized druglikeness, providing a robust framework for future laboratory synthesis and Ellman assay validation in the pursuit of disease-modifying Alzheimer's treatments.

RECOMMENDATIONS

Based on the computational evaluation of the substituted indole-carbamate derivatives, the following strategic recommendations are proposed to address the inherent limitations of *in silico* modeling and to advance the development of the lead candidates, 6d and 6e, toward therapeutic application.

1. Transition to In Vitro Validation

Transition to experimental assays, specifically the Ellman assay, to determine IC₅₀ values against AChE and BChE. This is essential to validate the predicted dual-site binding mechanism and resolve discrepancies between computational affinities and biological activity.

2. Molecular Dynamics (MD) Simulations

Conduct minimum 100 ns MD simulations to evaluate the temporal stability of ligand-protein complexes. This will ensure that the interactions of the halogenated derivative **6e** with the peripheral anionic site (PAS) remain robust under physiological conditions.

3. Advanced ADMET and Toxicity Testing

- **PAMPA-BBB assays** to quantify blood-brain barrier permeability.
- **MDR1-MDCK cell line studies** to verify non-P-gp substrate status and CNS residency.
- **Cyp450 inhibition assays** to assess potential drug-drug interactions.

4. Scaffold Refinement and SAR Expansion

Exploring bioisosteric replacements for the carbamate moiety may further enhance the metabolic stability of the Rivastigmine scaffold, potentially extending the half-life of the therapeutic lead.

5. Synthetic Feasibility Assessment

Finally, a rigorous assessment of the synthetic pathways for the indole-carbamate hybrids is required. Streamlining the modular synthesis of these Multi-Target Directed Ligands (MTDLs) will be crucial for scaling the project from computational design to laboratory synthesis and subsequent pre-clinical evaluation.

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<https://www.swissadme.ch/>
3. **ProTox 3.0 - Prediction Of Toxicity Of Chemicals**
virtual lab for the prediction of toxicities of small molecules.
The prediction of compound toxicities is an important part of the drug design development process. Computational toxicity estimations are not only faster than the determination of toxic doses in animals, but can also help to reduce the amount of animal experiments.
https://tox.charite.de/protox3/index.php?site=compound_input
4. **AdmetSAR 3.0 - useful and powerful tool for drug development and chemical risk assessment.**
Absorption, distribution, metabolism, excretion, and toxicity (ADMET) properties play a crucial role in drug development and chemical risk assessment.
<https://lmmd.ecust.edu.cn/admet3/predict.php>